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CHAPTER 10

FISCAL POLICY AND THE FEDERAL BUDGET

Introduction This chapter deals with fiscal policy and the federal budget. Fiscal policy deals with changes in government expenditures and/or taxes to achieve particular economic goals, such as low unemployment, stable prices, and economic growth. In the United States, the Congress and the president, together, fashion fiscal policy. We begin our discussion with some facts and figures about government expenditures and taxation, and we then go on to discuss the effect of fiscal policy on the economy.

THE FEDERAL BUDGET

The federal budget is composed of two, not necessarily equal, parts: government expenditures and tax revenues. You are familiar with the term *government purchases* from earlier chapters. Government expenditures—sometimes simply called government spending—are not the same as government purchases. Government expenditures include government purchases and (government) transfer payments.[1]

Government Expenditures

In 2007, the federal government spent $2.731 trillion—about 20 percent of GDP for that year. The following table shows government spending as a percentage of GDP in a few other years.

1. Remember from an earlier chapter that government purchases are the purchases of goods and services by government at all levels. Transfer payments are payments to persons that are not made in return for goods and services currently supplied, such as Social Security payments. In this chapter, the terms *government expenditures, government spending, government purchases,* and *transfer payments* all refer to the *federal* government.

Year	Government Spending as a Percentage of GDP
2000	18.4
2001	18.5
2002	19.4
2003	19.9
2004	19.9
2005	20.2
2006	20.3
2007	20.0
2008	20.4 (projected)
2009	20.3 (projected)

The bulk of the $2.731 trillion in government spending in 2007 was spent on four programs: national defense, Social Security, Medicare, and Medicaid. These four programs together accounted for 63.6 percent of all government spending in 2007. The following table shows the actual dollar amounts spent in various spending program categories.

Spending Program Category	Billions of Dollars
National Defense	$530
Social Security	577
Medicare	440
Medicaid	191
Other Programs and Activities	741
Net Interest on the Public Debt	252

Government Tax Revenues

The federal government imposes taxes and fees that generate revenue. In 2007, government revenues totaled $2.568 trillion. This was 18.8 percent of GDP for the year. The following table shows government tax revenues as a percentage of GDP in a few other years.

Year	Government Tax Revenues as a Percentage of GDP
2000	20.9
2001	19.8
2002	17.9
2003	16.5
2004	16.3
2005	17.5
2006	18.4
2007	18.8
2008	17.9 (projected)
2009	18.9 (projected)

The bulk of government tax revenues comes from three taxes: the individual income tax, the corporate income tax, and Social Security taxes. These three taxes accounted for 93.6 percent of total government tax revenues in 2007. The following table shows the

actual dollar amount raised in tax revenue by each tax, and the tax revenue for each tax as a percentage of GDP.

Tax	Billions of Dollars	Percentage of 2007 GDP
Individual Income Tax	$1,163	8.5
Corporate Income Tax	370	2.7
Social Security Taxes	870	6.3
Other	163	1.2

You can see from these numbers that the individual income tax is a large portion of the government tax revenue pie. Let's look at this tax in more detail.

INCOME TAX STRUCTURES An income tax structure can be progressive, proportional, or regressive. Under a **progressive income tax**, the tax rate increases as a person's taxable income level rises. To illustrate, suppose Davidson pays taxes at the rate of 15 percent on a taxable income of $20,000. When his (taxable) income rises to, say, $30,000, he pays at a rate of 28 percent. And when his income rises to, say, $55,000, he pays at a rate of 31 percent. A progressive income tax is usually capped at some rate. Currently, the U.S. income tax structure is progressive, with six (marginal) tax rates, ranging from a low of 10 percent to a high (or cap) of 35 percent.

Progressive Income Tax
An income tax system in which one's tax rate rises as one's taxable income rises (up to some point).

Under a **proportional income tax**, the same tax rate is used for all income levels. A proportional income tax is sometimes referred to as a *flat tax.* For example, if Kuan's taxable income is $10,000, she pays taxes at a rate of 10 percent; if her taxable income rises to $100,000, she still pays at a rate of 10 percent.

Proportional Income Tax
An income tax system in which a person's tax rate is the same no matter what his or her taxable income is.

Under a **regressive income tax**, the tax rate decreases as a person's taxable income level rises. For example, Lowenstein's tax rate is 10 percent when her taxable income is $10,000 and 8 percent when her taxable income rises to $20,000.

Regressive Income Tax
An income tax system in which a person's tax rate declines as his or her taxable income rises.

See Exhibit 1 for a review of the three income tax structures.

Finding ECONOMICS

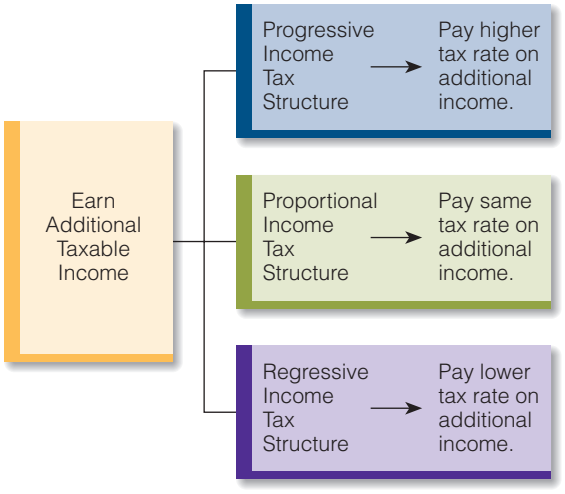
A Presidential Candidate Speaks

A presidential candidate is speaking before a large group in Des Moines, Iowa. He has just called the current income tax structure unfair. Someone from the crowd asks him what he means by "unfair." The candidate says that individuals shouldn't pay a higher tax rate just because they earn a higher income. We should all pay the same tax rate, he argues. Where is the economics? Obviously the candidate favors a proportional income tax (i.e., a flat tax). Under both a progressive and a regressive income tax structure, individuals pay different tax rates at different taxable income levels. Only with a proportional income tax do individuals pay the same tax rate no matter what their taxable income is.

exhibit 1

Three Income Tax Structures

The three income tax structures outlined are the progressive, proportional, and regressive.



WHO PAYS THE INCOME TAX? Economists often look at the tax situations of different income groups. For example, in 2005, the top 1 percent of income earners in the United States earned 21.20 percent of the total income earned that year and

TWO PLUMBERS, NEW YEAR'S EVE, AND PROGRESSIVE TAXATION

Many people believe that if two people do the same job, they should be paid the same dollar amount. This notion of equal pay for equal work often arises in discussions about the jobs performed by men and women. In other words, if a man and a woman do the same job, many people say that they should be paid the same dollar amount.

Sometimes the question of equal pay for equal work is extended to equal after-tax pay for equal work. That is, if two people do the same job, then they should earn the same after-tax income. However, a progressive income tax structure sometimes makes this impossible.

To illustrate, suppose under a progressive income tax structure, a person who earns between $50,000 and $60,000 pays income tax at a tax rate of 20 percent. For every dollar earned over $60,000 but under $70,000, a person pays at a tax rate of 30 percent.

Consider two plumbers, Smith and Jones. By December 30, Jones has earned $58,000 for the year, and Smith has earned $60,000. Each is asked to do the same kind of plumbing job on December 31, New Year's Eve. Each plumber charges $1,000 for the job. So Jones and Smith receive equal pay for equal work.

The after-tax income that each receives for the job makes for a different story. On the additional $1,000 that Jones earns, she pays at a tax rate of 20 percent. So she pays $200 in taxes and gets to keep $800 in after-tax income. Smith, on the other hand, now has an annual income of $61,000 and thus falls into a higher marginal tax bracket. He pays at a tax rate of 30 percent on the additional $1,000. So Smith pays $300 in taxes and has $700 in after-tax income. Smith does the same job as Jones but earns only $700 in after-tax pay, whereas Jones earns $800 in after-tax pay.

Our conclusion: The progressive income tax structure can turn equal pay for equal work into unequal after-tax pay for equal work. Stated differently, a person can be in favor of progressive income taxes or equal after-tax pay for equal work but not both. Sometimes, it is a matter of one or the other.

paid 39.38 percent of the total federal income taxes. The following data show the income and taxes for various income groups in 2005:

Income Group	Group's Share of Total Income	Group's Share of Federal Income Taxes
Top 1%	21.20	39.38
Top 5%	35.75	59.67
Top 10%	46.44	70.30
Top 25%	67.52	85.99
Top 50%	87.17	96.93
Bottom 50%	12.83	3.07

Common Misconceptions

About the Rich and Taxes

It isn't uncommon to hear people say the rich do not pay a high percentage of income taxes in the United States. Often, many of these people do not know exactly what percentages different income groups pay in federal income taxes. If we define the rich as those in the top 1 percent of income earners, then we see that in 2005, they paid 39.38 percent of all federal income taxes. Or consider this: From looking at the income and tax data, we see that the bottom 95 percent of income earners paid 40.33 percent (100.00 – 59.67) of all federal income taxes in 2005. This percentage is

(continued)

Common Misconceptions (continued)

very close to 39.38 percent, which is the percentage of federal income taxes paid by the top 1 percent of income earners. In other words, in 2005 the top 1 percent of income earners paid approximately the same percentage of income taxes as the bottom 95 percent. By the way, you had to earn more than $364,657 to be in the top 1 percent of income earners in 2005.

Budget Deficit, Surplus, or Balance

If government expenditures are greater than tax revenues, the federal government runs a **budget deficit**. If tax revenues are greater than government expenditures, the federal government runs a **budget surplus**. If government expenditures equal tax revenues, the federal government runs a **balanced budget**.

Budget Deficit
Government expenditures greater than tax revenues.

Budget Surplus
Tax revenues greater than government expenditures.

Balanced Budget
Government expenditures equal to tax revenues.

In 2007, government expenditures were $2.731 trillion, and tax revenues were $2.568 trillion; so the federal government ran a budget deficit that year of $163 billion. Budget deficits are projected for upcoming years. For the period 2008 to 2011, the government is projected to run budget deficits. On July 29, 2008, the Congressional Budget Office estimated the federal budget deficit for 2008 at $422 billion.

If the government spends more than its tax revenue and thus runs a budget deficit, where does it get the money to finance the deficit? In other words, if the government spends $100 and only has $70 in taxes, where does it get the $30 difference? The answer is that the federal government—actually the U.S. Treasury—borrows the $30; that is, it finances the budget deficit with borrowed funds.

Structural and Cyclical Deficits

Suppose the budget is currently balanced, and then Real GDP in the economy drops. As Real GDP drops, the tax base of the economy falls, and, if tax rates are held constant, tax revenues will fall. Also as a result of the decline in Real GDP, transfer payments (e.g., unemployment compensation) will rise. Thus, government expenditures will rise, and tax revenues will fall. As a result, a balanced budget turns into a budget deficit. This budget deficit results from the downturn in economic activity, not from any current spending and taxing decisions by the government.

Economists use the term **cyclical deficit** to refer to the part of the budget deficit that is a result of a downturn in economic activity. The remainder of the deficit—or the part of the deficit that would exist if the economy were operating at full employment—is called the **structural deficit**. In other words,

Cyclical Deficit
The part of the budget deficit that is a result of a downturn in economic activity.

Structural Deficit
The part of the budget deficit that would exist even if the economy were operating at full employment.

$$\text{Total budget deficit} = \text{Structural deficit} + \text{Cyclical deficit}$$

To illustrate, suppose the economy is in a recessionary gap, government expenditures are currently $2.3 trillion, and tax revenues are $2.0 trillion. Thus, the (total) budget deficit is $300 billion. Economists estimate what government expenditures and tax revenues would be if the economy were operating at full employment. Assume they estimate that government expenditures would be only $2.2 trillion and that tax revenues would be $2.1 trillion. The structural deficit—the deficit that would exist at full employment—is therefore $100 billion. The cyclical deficit—the part of the budget deficit that is a result of economic downturn—is $200 billion.

The Public Debt

A budget deficit occurs when government expenditures are greater than tax revenues for a *single year.* The **public debt**, which is sometimes called the federal or national debt, is the *total* amount the federal government owes its creditors. Some of this debt is held by agencies

Public Debt
The total amount that the federal government owes its creditors.

Q&A: GOVERNMENT SPENDING AND TAXES

Economics is not all about concepts, theories, and policies. Sometimes it is about facts and figures. A complete knowledge of the economic scene requires us to know some of those facts and figures, which are presented here as the answer to questions that individuals often ask.

What are the two largest federal taxes that U.S. households pay?

The Social Security tax and the federal income tax. The average household paid $7,069 in Social Security taxes in 2004 and $7,062 in federal income taxes. If we add all federal taxes, the average household paid a total of $17,338 in federal taxes in 2004.

What are the three largest state and local taxes that U.S. households pay?

Property taxes, general sales taxes, and individual income taxes. In 2004, the average household paid $2,906 in property taxes, $2,240 in general sales taxes, and $1,984 in income taxes. If we add all state and local taxes, the average household paid a total of $9,400.

How much do U.S. households pay in taxes?

The total dollar amount in 2004 (including federal, state, and local taxes) was $3 trillion. This was an average of $26,778 per household.

Do different income groups in the United States pay the same dollar amount in taxes?

No. In 2004, the average household in the bottom 20 percent of income-earning households paid $4,325 in taxes; the average household in the top 20 percent of income-earning households paid $81,933 in taxes. Households in the so-called middle class (the middle 20 percent of income-earning households) paid $21,194 in taxes.

Do low-income and high-income households alike pay approximately the same percentage of their taxes to the federal government as they do to the state and local governments?

No. The lowest-earning households (bottom 20 percent) pay more in state and local taxes than in federal taxes. The highest-earning households (top 20 percent) pay more in federal taxes than in state and local taxes. Here are the percentages: The lowest-earning households pay 39 cents out of each tax dollar to the federal government and 61 cents out of each tax dollar to the state and local governments. The highest-earning households pay 70 cents out of each tax dollar to the federal government and 30 cents to the state and local governments.

We've talked about taxes, but not about government spending. Is there any information on how much different people receive in government spending benefits compared to how much they pay in taxes?

Yes. In 2004, the bottom 20 percent of income-earning households paid an average of $2,642 in state and local taxes and received an average of $10,650 in (state and local) government spending benefits. In fact, the bottom 60 percent of income-earning households received more in government spending benefits than they paid in taxes. The top 40 percent of income-earning households paid more (on average) in state and local taxes than they received in state and local government spending. For example, the top 20 percent paid an average of $24,421 in state and local taxes and received an average of $14,911 in spending benefits.

What about federal taxes and federal government spending benefits? Do some households pay more in taxes than they receive in spending benefits?

Yes. Some households pay more in federal taxes than they receive in federal government spending benefits, and some households pay less in federal taxes than they receive in spending benefits. To illustrate, the top 20 percent of income-earning households paid an average of $57,512 in federal taxes in 2004 and received an average of $18,573 in spending benefits. The bottom 20 percent of households paid an average of $1,684 in federal taxes and received an average of $24,860 in spending benefits. Overall, the bottom 60 percent of households receive a greater dollar worth of spending benefits than they pay in taxes, and the top 40 percent pay a greater dollar amount in taxes than they receive in spending benefits.

If I receive something from the government that costs the government $10, does it follow that I receive at least $10 worth of benefits from the good or service?

No. To illustrate, a middle school student might be receiving a lunch at school that is paid for by the government, and it might cost the government $3 to provide the lunch. However, the middle school student does not necessarily value the benefits of the lunch at $3. The value of the benefits (the student receives from the lunch) could be either much higher or much lower than $3. It would be no different than my giving you a watch that I paid $100 for. You might not value the watch at $100. In other words, you might tell me that you receive only $40 worth of benefits from the watch (and that you would have never paid $100 for the watch if it were up to you).

of the United States government—one entity in the government owes it to another. The remainder of the debt is held by the public and is referred to either as *public debt held by the public* or as *net public debt*. The public debt was $9.5 trillion on July 29, 2008. The public debt held by the public was $5.3 trillion. You can find the current public debt on the Bureau of the Public Debt website at http://www.publicdebt.treas.gov/opd/opd.htm. The public debt was at its lowest on January 1, 1835, totaling $33,733.05 on that day.

SELF-TEST

(Answers to Self-Test questions are in the Self-Test Appendix.)

1. Explain the differences among progressive, proportional, and regressive income tax structures.
2. What percentage of all income taxes was paid by the top 5 percent of income earners in 2005? What percentage of total income did this income group receive in 2005?
3. What three taxes account for the bulk of federal tax revenues?
4. What is the cyclical budget deficit?

FISCAL POLICY

As explained in the last chapter, some economists believe that the economy is inherently unstable. These economists argue that government should play a role in managing the economy because the economy can get stuck in a recessionary gap. They believe government should try to move the economy out of the recessionary gap and toward Natural Real GDP.

One of the major ways government can influence the economy is through its *fiscal policy*. **Fiscal policy** consists of changes in government expenditures and/or taxes to achieve particular economic goals, such as low unemployment, price stability, and economic growth. We discuss fiscal policy in the following sections.

Fiscal Policy
Changes in government expenditures and/or taxes to achieve economic goals, such as low unemployment, stable prices, and economic growth.

Some Relevant Fiscal Policy Terms

Expansionary fiscal policy consists of increases in government expenditures and/or decreases in taxes to achieve macroeconomic goals. **Contractionary fiscal policy** is implemented through decreases in government expenditures and/or increases in taxes to achieve these goals.

Expansionary fiscal policy: Government expenditures up and/or taxes down

Contractionary fiscal policy: Government expenditures down and/or taxes up

When deliberate government actions bring about changes in its expenditures and taxes, fiscal policy is said to be *discretionary*. For example, if Congress decides to increase government spending by, say, $10 billion in an attempt to lower the unemployment rate, this is an act of **discretionary fiscal policy**. In contrast, a change in either government expenditures or in taxes that occurs automatically in response to economic events is referred to as **automatic fiscal policy**. To illustrate, suppose Real GDP in the economy turns down, causing more people to become unemployed, and, as a result, automatically receive unemployment benefits. These added unemployment benefits automatically boost government spending.

Expansionary Fiscal Policy
Increases in government expenditures and/or decreases in taxes to achieve particular economic goals.

Contractionary Fiscal Policy
Decreases in government expenditures and/or increases in taxes to achieve economic goals.

Discretionary Fiscal Policy
Deliberate changes of government expenditures and/or taxes to achieve economic goals.

Automatic Fiscal Policy
Changes in government expenditures and/or taxes that occur automatically without (additional) congressional action.

Two Important Notes

In your study of this chapter, keep in mind the following two important points:

1. In this chapter, we deal only with *discretionary fiscal policy*. In other words, we consider deliberate actions on the part of policy makers to affect the economy through changes in government spending and/or taxes.

2. We assume that any change in government spending is due to a change in government purchases, not to a change in transfer payments. Stated differently, we assume that transfer payments are constant so that changes in government spending are a reflection only of changes in government purchases.

Demand-Side Fiscal Policy

Fiscal policy can affect the demand side of the economy, that is, aggregate demand. This section focuses on how government spending and taxes can affect aggregate demand.

Shifting the Aggregate Demand Curve

How do changes in government purchases (*G*) and taxes (*T*) affect aggregate demand? Recall that a change in consumption, investment, government purchases, or net exports can change aggregate demand and therefore shift the *AD* curve. For example, an increase in government purchases (*G*) increases aggregate demand and shifts the *AD* curve to the right. A decrease in *G* decreases aggregate demand and shifts the *AD* curve to the left.[2]

A change in taxes (*T*) can affect consumption, investment, or both, and it therefore can affect aggregate demand. For example, a decrease in income taxes increases disposable (after-tax) income, which permits individuals to increase their consumption. As consumption rises, the *AD* curve shifts to the right. An increase in taxes decreases disposable income, lowers consumption, and shifts the *AD* curve to the left.

Fiscal Policy: Keynesian Perspective (Economy Is Not Self-Regulating)

The model of the economy in Exhibit 2(a) shows a downward-sloping *AD* curve and an upward-sloping *SRAS* curve. As you can see, the economy is initially in a recessionary gap at point 1. Aggregate demand is too low to move the economy to equilibrium at the Natural Real GDP level. The Keynesian *perspective* of the economy here is that the economy is not self-regulating. So the Keynesian *prescription* is to enact expansionary fiscal policy measures (an increase in government purchases or a decrease in taxes) to shift the aggregate demand curve rightward from AD_1 to AD_2 and to move the economy to the Natural Real GDP level at point 2.

At this point, the question might be, why not simply wait for the short-run aggregate supply curve to shift rightward and intersect the aggregate demand curve at point 2′? Again, the Keynesians usually respond that the economy is not self-regulating. They argue that either (1) the economy is stuck at point 1 and won't move naturally to point 2′—perhaps because wage rates won't fall—or (2) the short-run aggregate supply curve takes too long to shift rightward, and in the interim we must deal with the high cost of unemployment and a lower level of Real GDP. In Exhibit 2(b), the economy is initially in an inflationary gap at point 1. In this situation, Keynesians are likely to propose a contractionary fiscal measure (a decrease in government purchases or an increase in taxes) to shift the aggregate demand curve leftward from AD_1 to AD_2 and move the economy to point 2.

In Exhibit 2, fiscal policy has worked as intended. In (a), the economy was in a recessionary gap, and expansionary fiscal policy eliminated the recessionary gap. In (b), the economy was in an inflationary gap, and contractionary fiscal policy eliminated the inflationary gap. In (a) and (b), fiscal policy is at its best and working as intended.

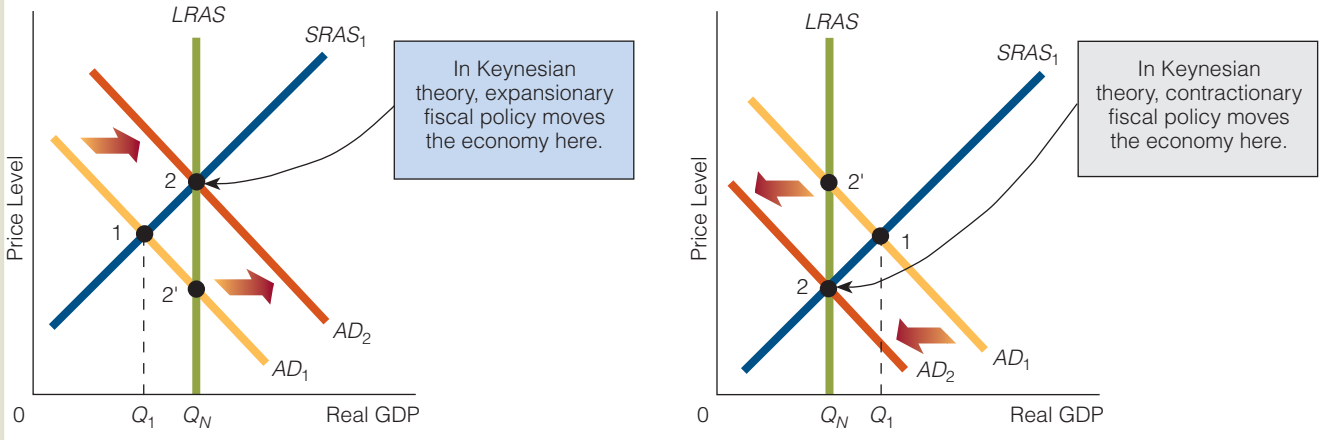
2. Later in this chapter, when we discuss crowding out, we question the effect of an increase in government purchases on aggregate demand.

exhibit 2

Fiscal Policy in Keynesian Theory: Ridding the Economy of Recessionary and Inflationary Gaps

(a) In Keynesian theory, expansionary fiscal policy eliminates a recessionary gap. Increased government purchases, decreased taxes, or both lead to a rightward shift in the aggregate demand curve from AD_1 to AD_2, restoring the economy to the natural level of Real GDP, Q_N.

(b) Contractionary fiscal policy is used to eliminate an inflationary gap. Decreased government purchases, increased taxes, or both lead to a leftward shift in the aggregate demand curve from AD_1 to AD_2, restoring the economy to the natural level of Real GDP, Q_N.



(a) Expansionary Fiscal Policy for a Recessionary Gap

(b) Contractionary Fiscal Policy for an Inflationary Gap

Common Misconceptions

About Fiscal Policy

In 1962, John F. Kennedy was president of the United States, and Walter Heller was one of Kennedy's economic advisors. Heller told the president that the economy needed a tax cut (a form of expansionary fiscal policy) to keep it from sputtering. In December, in a speech before the Economic Club of New York, President Kennedy said, "An economy hampered by restrictive tax rates will never produce enough revenue to balance our budget just as it will never produce enough jobs or enough profits."

Then in January 1963, he said, "It has become increasingly clear that the largest single barrier to full employment . . . and to a higher rate of economic growth is the unrealistically heavy drag of federal income taxes on private purchasing power, initiative and incentive." Kennedy proposed expansionary fiscal policy—in the form of a tax cut—to raise economic growth and lower the unemployment rate. He proposed lowering the top individual income tax rate, the bottom individual income tax rate, the corporate income tax, and the capital gains tax. He was assassinated in Dallas before Congress passed his tax program, but Congress did pass it. What was the result?

When the tax bill passed in 1964, the unemployment rate was 5.2 percent; in 1965, it was down to 4.5 percent; in 1966, it was down further to 3.8 percent. The tax cut is widely credited with bringing the unemployment rate down. As for economic growth, when the tax cut was passed in 1964, it was 5.8 percent; one year later, in 1965, the growth rate was up to 6.4 percent; and in 1966, the growth rate was even higher, at 6.6 percent. Again, the tax cut received much of the credit for stimulating economic growth.

macro*theme* ➜ In an earlier chapter, we said that economists don't always agree that economic policy is effective at, say, removing an economy from a recessionary or inflationary gap. Specifically, some economists say that fiscal policy is effective, whereas others say that it is ineffective. You have just heard from the economists who say fiscal policy is effective. Now we turn to those who say it is not.

Crowding Out: Questioning Expansionary Fiscal Policy

Crowding Out
The decrease in private expenditures that occurs as a consequence of increased government spending or the financing needs of a budget deficit.

Not all economists believe that fiscal policy works as we have just described. Some economists bring up the subject of *crowding out.* **Crowding out** is a decrease in private expenditures (consumption, investment, etc.) as a consequence of increased government spending or the financing needs of a budget deficit.

Crowding out can be direct or indirect, as described in these two examples:

1. *Direct effect.* The government spends more on public libraries, and individuals buy fewer books at bookstores.[3]
2. *Indirect effect.* The government spends more on social programs and defense without increasing taxes; as a result, the size of the budget deficit increases. Consequently, the government must borrow more funds to finance the larger deficit. This increase in borrowing causes the demand for credit (i.e., the demand for loanable funds) to rise, which in turn causes the interest rate to rise. As a result, investment drops. More government spending indirectly leads to less investment spending.

TYPES OF CROWDING OUT In our first example, the government spends more on public libraries. To be specific, let's say that the government spends $2 billion more on public libraries and that consumers choose to spend not $1 less on books at bookstores. Obviously, then, there is no crowding out, or *zero crowding out*.

Now suppose that, after the government has spent $2 billion more on public libraries, consumers choose to spend $2 billion less on books at bookstores. Obviously, crowding out exists, and the degree of crowding out is dollar for dollar. When $1 of government spending offsets $1 of private spending, **complete crowding out** is said to exist.

Complete Crowding Out
A decrease in one or more components of private spending that completely offsets the increase in government spending.

Incomplete Crowding Out
The decrease in one or more components of private spending that only partially offsets the increase in government spending.

Finally, suppose that, after the government has spent $2 billion more on public libraries, consumers spend $1.2 billion less on books at bookstores. Again, there is crowding out, but it is not dollar for dollar, not complete crowding out. In this case, incomplete crowding out exists. **Incomplete crowding out** occurs when the decrease in one or more components of private spending only partially offsets the increase in government spending.

The following table summarizes the different types of crowding out.

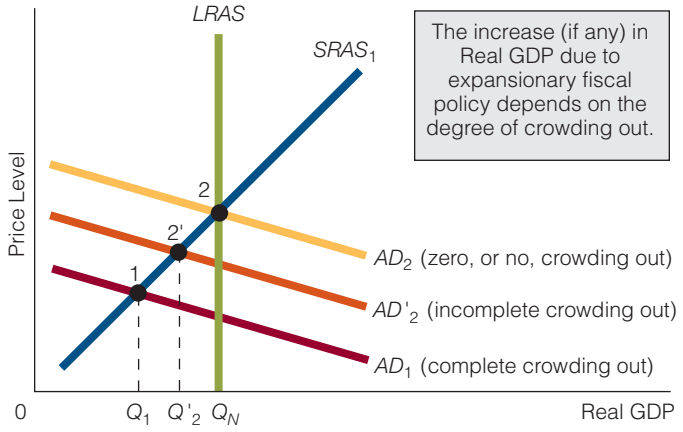
Type of Crowding Out	Example
Zero crowding out (sometimes called "no crowding out")	Government spends $2 billion more, and private sector spending stays constant.
Complete crowding out	Government spends $2 billion more, and private sector spends $2 billion less.
Incomplete crowding out	Government spends $2 billion more, and private sector spends $1.2 billion less.

3. We are not saying that, for example, if the government spends more on public libraries, individuals will necessarily buy fewer books at bookstores; rather, if they do, this would be an example of crowding out. The same holds for example 2.

exhibit 3

Zero (No), Incomplete, and Complete Crowding Out

The exhibit shows the effects of zero, incomplete, and complete crowding out in the *AD-AS* framework. Starting at point 1, expansionary fiscal policy shifts the aggregate demand curve to AD_2 and moves the economy to point 2 and Q_N. The Keynesian theory that predicts this outcome assumes zero, or no, crowding out; an increase in, say, government spending does not reduce private expenditures. With incomplete crowding out, an increase in government spending causes private expenditures to decrease by less than the increase in government spending. The net result is a shift in the aggregate demand curve to AD'_2. The economy moves to point 2′ and Q'_2. With complete crowding out, an increase in government spending is completely offset by a decrease in private expenditures, and the net result is that aggregate demand does not increase at all. The economy remains at point 1 and Q_1.



GRAPHICAL REPRESENTATION OF CROWDING OUT If *complete* or *incomplete crowding out* occurs, then expansionary fiscal policy will have less impact on aggregate demand and Real GDP than Keynesian theory predicts. Let's look at the graphical representation of crowding out.

Exhibit 3 illustrates the consequences of complete and incomplete crowding out. For comparison, the exhibit also includes the case of zero crowding out in Keynesian theory. As shown in Exhibit 3, keep in mind the three possibilities concerning crowding out:

- Zero crowding out (no crowding out)
- Incomplete crowding out
- Complete crowding out

In Exhibit 3, the economy is initially at point 1, with Real GDP at Q_1. In Keynesian theory, expansionary fiscal policy shifts the aggregate demand curve to AD_2 and moves the economy to point 2. Among other things, the implicit assumption is that there is zero crowding out (no crowding out). Notice that Real GDP has increased from Q_1 to Q_N. It follows that the unemployment rate will fall from its level at Q_1 to a lower level at Q_N. Summary: If there is no crowding out, expansionary fiscal policy increases Real GDP and lowers the unemployment rate.

With incomplete crowding out, the aggregate demand curve shifts (on net) only to AD'_2 because a fall in private expenditures *partially offsets* the initial stimulus in aggregate demand due to increased government spending. The economy moves to point 2′. Notice that Real GDP has increased from Q_1 to Q'_2. It follows that the unemployment rate will fall from what it was at Q_1 to what it is at Q'_2. Also notice that the changes in both Real GDP and the unemployment rate are smaller, with incomplete crowding out than they are with zero crowding out. Summary: Given incomplete crowding out, expansionary fiscal policy increases Real GDP and lowers the unemployment rate but not as much as if there is zero crowding out.

MOVIE CROWDING OUT: THE CASE OF *THE DARK KNIGHT*

The blockbuster movie *The Dark Knight* made its U.S. premiere release on July 18, 2008. In its first three days at the theaters, it took in $158 million in gross receipts.

WARNER BROS/DC COMICS/THE KOBAL COLLECTION

As movie releases go, $158 million (in three days) is an extraordinarily large dollar amount. Some people said this dollar amount indicated that the public was spending more money on going to the movies.

But is this statement necessarily true? Certainly, it doesn't have to be. There may be such a thing as movie crowding out. To illustrate, we assume that the moviegoing public spends $70 million each weekend on ten movies. Let's say this dollar amount is evenly distributed across all ten movies so that each movie earns $7 million. A blockbuster movie may simply have a larger share of the $70 million pie. The blockbuster may earn, say, $20 million, and the nine remaining movies evenly divide the remaining $50 million. In other words, spending on the blockbuster comes at the expense of other movies on a dollar-for-dollar basis. Blockbuster spending crowds out nonblockbuster spending in much the same way as government spending can crowd out household spending.

Movie crowding out could also work another way. Perhaps because of the blockbuster, total spending on movies rises when the blockbuster is released; that is, the moviegoing public increases the amount spent on movies in the first few weekends after a blockbuster is released. Thus, spending may rise to, say, $100 million each weekend for three consecutive weekends after the release of a blockbuster. But then what we might call the blockbuster effect fades away, and spending on movies falls below the usual $70 million per weekend. It may fall to, say, $50 million per weekend for a few weekends. Blockbuster spending has still crowded out nonblockbuster spending, but not as quickly as in the first case.

There is also a related issue. Perhaps a blockbuster doesn't crowd out other movie spending but does crowd out nonmovie spending. To illustrate, suppose that, because of a blockbuster, spending on movies actually rises (over a year). The new average goes from $70 million each weekend to, say, $80 million. But because people are spending more on movies, they spend less on other things. In other words, movie spending crowds out nonmovie spending. People spend less on books, restaurant meals, clothes, and the like. One sector of the economy (the movie sector) expands as another contracts.

In the case of complete crowding out, a fall in private expenditures *completely offsets* the initial stimulus in aggregate demand due to increased government spending, and the aggregate demand curve does not move (on net) at all. Notice that Real GDP does not change, and neither does the unemployment rate. Summary: If there is complete crowding out, expansionary fiscal policy has no effect on the economy. The economy remains at point 1.

See Exhibit 4 for a summary flow chart of the different types of crowding out.

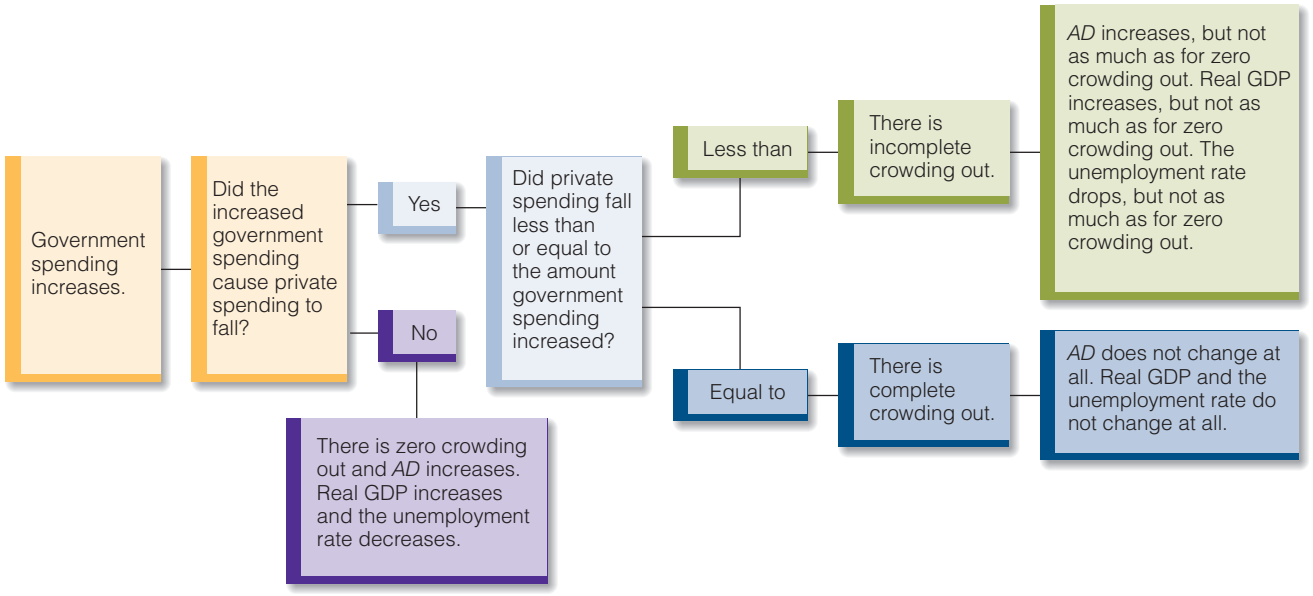
Thinking like An Economist

Policy Is Not Necessarily Effective

An ill person goes to the doctor and asks for medicine. The doctor prescribes the medicine, and the person goes home. After a few days, the medicine has not made the person well. The same can be sometimes said of certain types of economic policy. Keep in mind what we are and are not saying. We are not saying that economic policy is never effective; we are simply saying it is not necessarily effective. In our discussion of fiscal policy so far, crowding out is simply one reason fiscal policy may not be effective at times. Lags, which we discuss next, are another.

exhibit 4

Expansionary Fiscal Policy (Government Spending Increases), Crowding Out, and Changes in Real GDP and the Unemployment Rate



Lags and Fiscal Policy

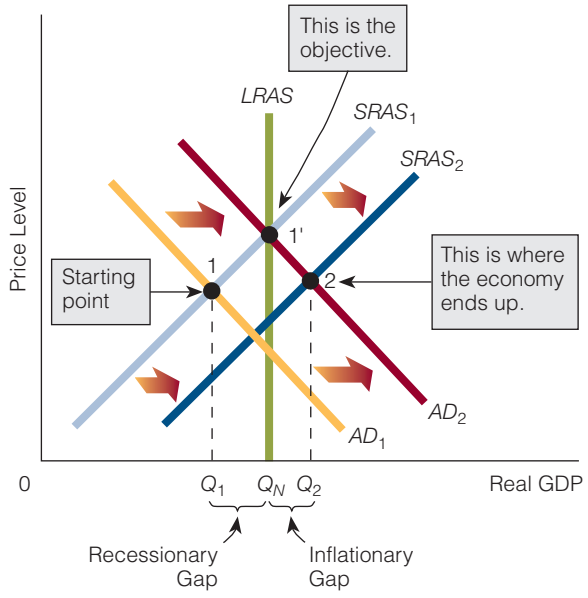
Suppose we proved beyond a shadow of a doubt that no (zero) crowding out is taking place. Should fiscal policy then be used to solve the problems of inflationary and recessionary gaps? Many economists would answer not necessarily. The reason is that *lags* exist. There are five types of lags:

1. *The data lag.* Policy makers are not aware of changes in the economy as soon as they happen. For example, if the economy turns down in January, the decline may not be apparent for two to three months.
2. *The wait-and-see lag.* After policy makers are aware of a downturn in economic activity, they rarely enact counteractive measures immediately. Instead, they usually adopt a relatively cautious wait-and-see attitude. They want to be sure that the observed events are not just short-run phenomena.
3. *The legislative lag.* After policy makers decide that some type of fiscal policy measure is required, Congress or the president has to propose the measure, build political support for it, and get it passed. The legislative lag can take many months.
4. *The transmission lag.* After enacted, a fiscal policy measure takes time to go into effect. For example, a discretionary expansionary fiscal policy measure mandating increased spending for public works projects requires construction companies to submit bids for the work, prepare designs, negotiate contracts, and so on.
5. *The effectiveness lag.* After a policy measure is actually implemented, it takes time to affect the economy. If government spending is increased on Monday, the aggregate demand curve does not shift rightward on Tuesday.

exhibit 5

Fiscal Policy May Destabilize the Economy

In this scenario, the *SRAS* curve is shifting rightward (healing the economy of its recessionary gap), but this information is unknown to policy makers. Policy makers implement expansionary fiscal policy, and the *AD* curve ends up intersecting $SRAS_2$ at point 2 instead of intersecting $SRAS_1$ at point 1′. Policy makers thereby move the economy into an inflationary gap, thus destabilizing the economy.



Taking these five lags together, some economists argue that discretionary fiscal policy is not likely to have the impact on the economy that policy makers hope for. By the time the full impact of the policy is felt, the economic problem it was designed to solve (1) may no longer exist, (2) may not exist to the degree it once did, or (3) may have changed altogether.

Exhibit 5 illustrates the effect of lags. Suppose the economy is currently in a recessionary gap at point 1. The recession is under way before government officials recognize it. After it is recognized, however, Congress and the president consider enacting expansionary fiscal policy in the hope of shifting the *AD* curve from AD_1 to AD_2 so that it will intersect the *SRAS* curve at point 1′, at Natural Real GDP.

In the interim, unknown to everybody, the economy is said to be healing, or regulating, itself: The *SRAS* curve is shifting to the right. Government officials don't see this change because it takes time to collect and analyze data about the economy.

Thinking that the economy is not healing itself or not healing itself quickly enough, the government enacts expansionary fiscal policy. In time, the *AD* curve shifts rightward. But by the time the increased demand is felt in the goods and services market, the *AD* curve intersects the *SRAS* curve at point 2. In short, the government has moved the economy from point 1 to point 2, not, as it had desired, from point 1 to point 1′. The government has moved the economy into an inflationary gap. Instead of stabilizing and moderating the ups and downs in economic activity (the business cycle), the government has intensified the fluctuations.

Crowding Out, Lags, and the Effectiveness of Fiscal Policy

Economists who believe that there is zero crowding out and that lags are insignificant conclude that fiscal policy is effective at moving the economy out of a recessionary gap. Economists who believe that crowding out is complete and/or that lags are significant conclude that fiscal policy is ineffective at moving the economy out of a recessionary gap.

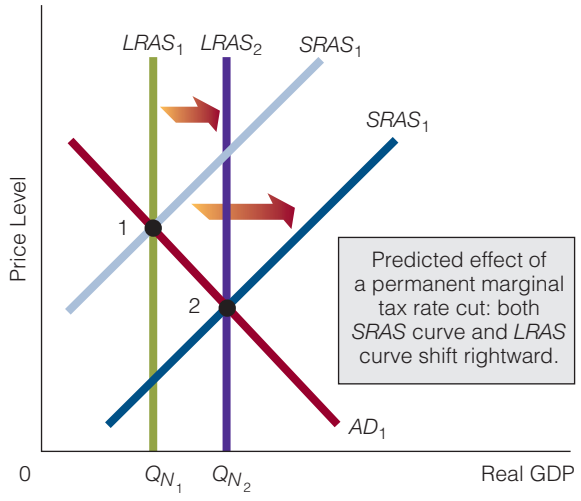
SELF-TEST

1. How does crowding out create questions about the effectiveness of expansionary demand-side fiscal policy? Give an example.
2. How might lags reduce the effectiveness of fiscal policy?
3. Give an example of the indirect effect of crowding out.

Supply-Side Fiscal Policy

Fiscal policy effects may be felt on the supply side as well as on the demand side of the economy. For example, a reduction in tax rates may alter an individual's incentive to work and produce, thus altering aggregate supply.

exhibit 6



The Predicted Effect of a Permanent Marginal Tax Rate Cut on Aggregate Supply

A cut in marginal tax rates increases the attractiveness of productive activity relative to leisure and tax-avoidance activities and shifts resources from the latter to the former, thus shifting rightward both the short-run and the long-run aggregate supply curves.

Marginal Tax Rates and Aggregate Supply

When fiscal policy measures affect tax rates, they may affect both aggregate supply and aggregate demand. Consider a reduction in an individual's marginal tax rate. The **marginal (income) tax rate** is equal to the change in a person's tax payment divided by the change in the person's taxable income.

Marginal (Income) Tax Rate
The change in a person's tax payment divided by the change in his or her taxable income: ΔTax payment/ΔTaxable income.

$$\text{Marginal tax rate} = \frac{\Delta\text{Tax payment}}{\Delta\text{Taxable income}}$$

For example, if Serena's taxable income increases by $1 and her tax payment increases by $0.28, her marginal tax rate is 28 percent; if her taxable income increases by $1 and her tax payment increases by $0.35, then her marginal tax rate is 35 percent.

All other things held constant, lower marginal tax rates increase the incentive to engage in productive activities (work) relative to leisure and tax-avoidance activities.[4] As resources shift from leisure to work, short-run aggregate supply increases. If the lower marginal tax rates are permanent and not simply a one-shot affair, most economists predict that not only will the short-run aggregate supply curve shift rightward, but the long-run aggregate supply curve will shift rightward too. Exhibit 6 illustrates the predicted effect of a permanent marginal tax rate cut on aggregate supply.

The Laffer Curve: Tax Rates and Tax Revenues

> High tax rates are followed by attempts of ingenious men to beat them as surely as snow is followed by little boys on sleds.
>
> —Arthur Okun, economist (1928–1980)

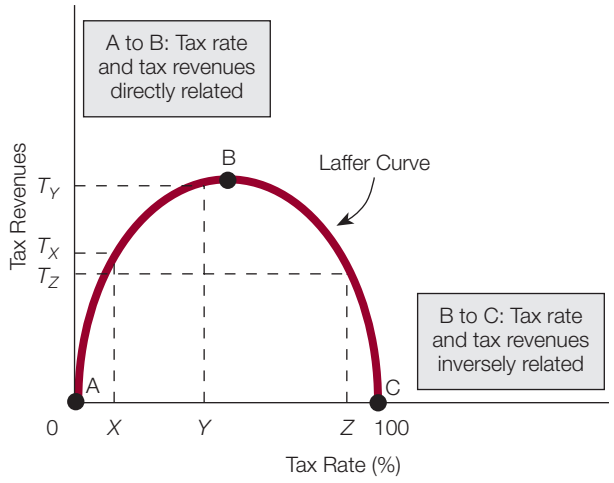
If (marginal) income tax rates are reduced, will income tax revenues increase or decrease? Most people think the answer is obvious: Lower tax rates mean lower tax revenues.

4. When marginal tax rates are lowered, two things happen: (1) Individuals will have more disposable income, and (2) the amount of money that individuals can earn (and keep) by working increases. As a result of the first effect, individuals will choose to work less. As a result of the second effect, individuals will choose to work more. Whether an individual works less or more on net depends on whether effect 1 is stronger than or weaker than effect 2. We have assumed that effect 2 is stronger than effect 1; so, as marginal tax rates decline, the net effect is that individuals work more.

exhibit 7

The Laffer Curve

When the tax rate is either 0 or 100 percent, tax revenues are zero. Starting from a zero tax rate, increases in tax rates first increase (region A to B) and then decrease (region B to C) tax revenues. Starting from a 100 percent tax rate, decreases in tax rates first increase tax revenues (region C to B) and then decrease tax revenues (region B to A). This suggests there is some tax rate that maximizes tax revenues.



Economist Arthur Laffer explained why this may not be the case. As the story is told, Laffer, while dining with a journalist at a restaurant in Washington, D.C., drew the curve in Exhibit 7 on a napkin. The curve came to be known as the **Laffer curve**. Laffer's objective was to explain the possible relationships between tax rates and tax revenues. In the exhibit, tax revenues are on the vertical axis, and tax rates are on the horizontal axis. Laffer made three major points about the curve:

1. Zero tax revenues will be collected at two (marginal) tax rates: 0 percent and 100 percent. Obviously, no tax revenues will be raised if the tax rate is zero, and if the tax rate is 100 percent, no one will work and earn income because the entire amount would be taxed away.
2. An increase in tax rates could cause tax revenues to increase. For example, an increase in tax rates from X percent to Y percent will increase tax revenues from T_X to T_Y.
3. A decrease in tax rates could cause tax revenues to increase. For example, a decrease in tax rates from Z percent to Y percent will increase tax revenues from T_Z to T_Y. This was the point that brought public attention to the Laffer curve.

Laffer Curve
The curve, named after Arthur Laffer, that shows the relationship between tax rates and tax revenues. According to the Laffer curve, as tax rates rise from zero, tax revenues rise, reach a maximum at some point, and then fall with further increases in tax rates.

Tax Base
In terms of income taxes, the total amount of taxable income.
Tax revenue = Tax base × (average) Tax rate.

How can an *increase* in tax rates and a *decrease* in tax rates at different times both increase tax revenues? This can happen because of the interrelationship of tax rates, the **tax base**, and tax revenues.

Tax revenues equal the tax base times the (average) tax rate:[5]

$$\text{Tax revenues} = \text{Tax base} \times \text{(average) Tax rate}$$

For example, a tax rate of 20 percent multiplied by a tax base of $100 billion generates $20 billion of tax revenues.

Obviously, tax revenues are a function of two variables: (1) the tax rate and (2) the tax base. Whether tax revenues increase or decrease as the average tax rate is lowered depends on whether the tax base expands by a greater or lesser percentage than the percentage reduction in the tax rate. Exhibit 8 illustrates the point. We start with a tax rate of 20 percent, a tax base of $100 billion, and tax revenues of $20 billion. We assume that as the tax rate is reduced, the tax base expands: The rationale is that individuals work more, invest more, enter into more trades, and shelter less income from taxes at lower tax rates.

However, the real question is *how much* does the tax base expand following the tax rate reduction? Suppose the tax rate in Exhibit 8 is reduced to 15 percent. In Case 1, the reduction increases the tax base to $120 billion: A 25 percent decrease in the tax rate (from 20 to 15 percent) causes a 20 percent increase in the tax base (from $100 billion

5. First, the average tax rate is equal to an individual's tax payment divided by his or her taxable income (tax payment/taxable income). Second, a lower average tax rate requires a lower marginal tax rate. This follows from the average-marginal rule, which states that if the marginal magnitude is below the average magnitude, then the average is pulled down; if the marginal is above the average, the average is pulled up. Simply put, if an individual pays less tax on an additional taxable dollar (which is evidence of a marginal tax rate reduction), then his or her average tax naturally falls.

exhibit 8

Tax Rates, the Tax Base, and Tax Revenues

Tax revenues equal the tax base times the (average) tax rate. If the percentage reduction in the tax rate is greater than the percentage increase in the tax base, tax revenues decrease (Case 1). If the percentage reduction in the tax rate is less than the percentage increase in the tax base, tax revenues increase (Case 2). All dollar amounts are in billions of dollars.

	(1) Tax Rate	(2) Tax Base	(3) Tax Revenues (1) × (2)	Summary
Start with:	20%	$100	$20	—
Case 1:	15	120	18	↓ Tax rate ↓ Tax revenues
Case 2:	15	150	22.5	↓ Tax rate ↑ Tax revenues

to $120 billion). Tax revenues drop to $18 billion. In Case 2, the tax base expands by 50 percent to $150 billion. Because the tax base increases by a greater percentage than the percentage decrease in the tax rate, tax revenues increase (to $22.5 billion).

Of course, either case is possible. In the Laffer curve, tax revenues increase if a tax rate reduction is made in the downward-sloping portion of the curve (between points B and C in Exhibit 8); tax revenues decrease following a tax rate reduction in the upward-sloping portion of the curve (between points A and B).

Thinking like An Economist

Incentives Matter

Contrast how economist Laffer thinks about a tax cut with the way the layperson thinks about it. The layperson probably believes that a reduction in tax rates will reduce tax revenues, focusing on the arithmetic of the situation. Laffer, however, focuses on the economic incentives. He asks what does a lower tax rate imply in terms of a person's incentive to engage in productive activity? How does a lower tax rate affect one's trade-off between work and leisure? The layperson likely sees only the arithmetic effect of a tax cut; the economist sees the incentive effect.

SELF-TEST

1. Give an arithmetic example to illustrate the difference between the marginal and average tax rates.
2. If income tax rates rise, will income tax revenues rise too?